STUDYLAB HUB · AP & COLLEGE GENERAL CHEMISTRY · UNIT 1 (CHAPTERS 1–2)

Unit 1: Atomic Structure & Properties

Moles & Molar Mass · Mass Spec · Empirical Formulas · Electron Configuration · PES · Periodic Trends

AP & College General Chemistry — Unit 1 (Chapters 1–2): Atomic Structure & Properties

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1. Unit 1 Roadmap (Chapters 1–2)

Topic	Core Formula / Principle	Exam Trap to Watch For
1.1 · Moles & Molar Mass	$n = \frac{m}{M} = \frac{N}{N_A}$, where $N_A = 6.022 \times 10^{23}\text{ mol}^{-1}$	Confusing **molecules** of a diatomic/polyatomic substance (e.g., O_2, CaCl_2) with total **atoms or ions** (1 mol CaCl_2 yields **3 mol ions**)
1.2 · Mass Spectroscopy of Elements	$\text{Average Atomic Mass} = \sum (f_i \times m_i)$	Never take a simple unweighted average of isotope masses; weight each peak by its relative abundance
1.3–1.4 · Elemental Composition & Empirical Formula	Mass $\% \to$ Moles $\to$ Divide by smallest mole value $\to$ Whole-number ratio	Rounding too early: a mole ratio of $1 : 1.33$ must be multiplied by 3 to give X_3Y_4, not rounded to $1:1$
1.5 · Atomic Structure & Electron Configuration	Aufbau, Pauli Exclusion, Hund's Rule; Coulomb's Law $F \propto \frac{q_1 q_2}{r^2}$	**Transition metal cations** lose valence **ns electrons BEFORE $(n-1)d$ electrons** (e.g., Fe^{2+} is $[\text{Ar}]\,3d^6$, *not* $[\text{Ar}]\,4s^2 3d^4$)
1.6 · Photoelectron Spectroscopy (PES)	$E_{\text{binding}} = h\nu - KE$; peak height $\propto$ number of electrons in subshell	Reading the x-axis backwards: PES binding energy often **decreases** from left to right (highest binding energy = $1s$ core electrons closest to nucleus)
1.7–1.8 · Periodic Trends & Valence Electrons	Effective nuclear charge (Z_{eff}) vs. occupied principal shells (n)	Stating "going right across a period increases trend X" as an explanation — **the trend is the observation, not the cause!** Always cite **Z_{eff} (more protons, same core shielding)** or **shell distance (n)**

2. Coulomb's Law: The Master Key to Every Unit 1 Question

Every atomic trend and PES question is governed by **Coulomb's Law**:

$$E_{\text{attraction}} \propto \frac{q_{\text{nucleus}} \cdot q_{\text{electron}}}{r}$$

1. **Across a Period (Left $\to$ Right)**:

- Protons (Z) increase while inner core electrons stay constant $\implies$ **Effective nuclear charge (Z_{eff}) increases**.
- Stronger attraction pulls the valence shell closer $\implies$ **Atomic radius decreases**, **First Ionization Energy (IE_1) increases**, and **Electronegativity increases**.

1. **Down a Group (Top $\to$ Bottom)**:

- Valence electrons occupy a higher principal energy level (n increases) with more inner core electron shielding $\implies$ distance r from the nucleus increases.
- Weaker attraction $\implies$ **Atomic radius increases**, **IE_1 decreases**, and **Electronegativity decreases**.

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3. Two Famous First Ionization Energy (IE_1) Exceptions

Anomaly	What Happens	Exact Orbital Explanation for FRQs
Group 2 vs. Group 13 (e.g., $\text{Be} > \text{B}$ or $\text{Mg} > \text{Al}$)	$IE_1(\text{B}) < IE_1(\text{Be})$ despite Boron having 1 more proton	Boron's outer electron is in a **$2p$ subshell** ($[\text{He}]2s^2 2p^1$), which is slightly higher in energy and partially shielded by the filled $2s^2$ electrons, making it easier to remove than a $2s$ electron from Be
Group 15 vs. Group 16 (e.g., $\text{N} > \text{O}$ or $\text{P} > \text{S}$)	$IE_1(\text{O}) < IE_1(\text{N})$ despite Oxygen having 1 more proton	Nitrogen ($2p^3$) has three unpaired electrons in separate $2p$ orbitals (Hund's rule). Oxygen ($2p^4$) forces a fourth electron to **pair up in one $2p$ orbital**, introducing **electron-electron repulsion** that makes that paired electron easier to remove

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4. Photoelectron Spectroscopy (PES) Rapid Decoder

- **X-Axis (Binding Energy)**: Measures the energy required to eject an electron from a specific subshell.
- **Highest Binding Energy Peak** = **$1s$ core electrons** (closest to the nucleus, smallest r, strongest Coulombic attraction).
- **Lowest Binding Energy Peak** = **Valence electrons** (furthest from the nucleus, highest n, shielded by core electrons).
- **Y-Axis (Relative Number of Electrons / Peak Height)**:
- If the $1s$ peak has height 2, then a $2p$ peak of height 6 is **3 times taller**, and a $3s^1$ peak (Sodium, Na) is **half as tall** as the $1s^2$ peak.
- **Comparing Two Elements' PES Spectra**:
- Because Oxygen ($Z=8$) has more protons than Nitrogen ($Z=7$), **every corresponding subshell peak ($1s, 2s, 2p$) in Oxygen appears at a higher binding energy** than in Nitrogen.

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5. Empirical & Molecular Formula + Hydrate Lab Calculations

Standard Empirical Formula Workflow

Given mass percentages (e.g., $40.0\%\text{ C}$, $6.71\%\text{ H}$, $53.29\%\text{ O}$):

1. Assume 100 g sample $\implies 40.0\text{ g C}, 6.71\text{ g H}, 53.3\text{ g O}$.
2. Convert to moles:

- $n_{\text{C}} = 40.0 / 12.01 = 3.33\text{ mol}$
- $n_{\text{H}} = 6.71 / 1.008 = 6.66\text{ mol}$
- $n_{\text{O}} = 53.29 / 16.00 = 3.33\text{ mol}$

1. Divide by smallest (3.33): $\text{C}_1\text{H}_2\text{O}_1 \implies$ Empirical formula = **CH_2O** ($M_{\text{emp}} = 30.03\text{ g/mol}$).
2. If experimental molar mass is 180.16 g/mol, multiplier $= 180.16 / 30.03 = 6 \implies$ Molecular formula = **$\text{C}_6\text{H}_{12}\text{O}_6$**.

Hydrate Heating Lab ($\text{CuSO}_4 \cdot x\text{H}_2\text{O}$)

- **Mass loss upon heating** = mass of evaporated H_2O.
- **Remaining solid mass** = mass of anhydrous salt (CuSO_4).
- **Common Lab Error Questions**:
- *If the hydrate is not heated long enough*: some water remains trapped $\implies$ calculated mass of H_2O lost is **too low** $\implies$ calculated value of x is **too low**.
- *If solid splatters out of the crucible*: lost solid is mistakenly counted as lost water $\implies$ calculated mass of H_2O is **too high** $\implies$ calculated value of x is **too high**.

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6. Worked Exam Practice (MCQs + Scored FRQ)

Practice Question 1 (Isoelectronic Ion Radii)

Rank the following isoelectronic species in order of increasing ionic/atomic radius: $\text{O}^{2-}, \text{F}^-, \text{Ne}, \text{Na}^+, \text{Mg}^{2+}$. Justify using Coulomb's Law.

- **Answer**: **$\text{Mg}^{2+} < \text{Na}^+ < \text{Ne} < \text{F}^- < \text{O}^{2-}$**.
- **Justification**: All five species have the exact same electron configuration ($1s^2 2s^2 2p^6$, 10 electrons) and identical core shielding. However, nuclear charge increases from O^{2-} (8 protons) to Mg^{2+} (12 protons). By Coulomb's Law, the 12 protons in Mg^{2+} exert the strongest attractive force on the 10 electrons, pulling the $n=2$ shell closest to the nucleus.

Practice Question 2 (Successive Ionization Energies)

Element X in Period 3 has the following successive ionization energies (in kJ/mol): $IE_1 = 578$, $IE_2 = 1,817$, $IE_3 = 2,745$, $IE_4 = 11,577$, $IE_5 = 14,842$. Identify Element X and write the electron configuration of its most common stable ion.

- **Answer**: The massive jump occurs between IE_3 ($2,745$) and IE_4 ($11,577$), indicating that the 4th electron is removed from an inner core shell ($n=2$) after 3 valence electrons are removed. In Period 3, the element with 3 valence electrons is **Aluminum (Al)**. Its stable ion is Al^{3+} with electron configuration **$1s^2 2s^2 2p^6$**.